| | | | | | | |
|---|---|---|---|---|---|---|
| **2** | **(a)** | | $(x \pm 6)^2$ and $(y \pm 4)^2$<br><br>$r$ = 7 not from wrong working | **M1**<br><br>**A1** | **1.1**<br><br>**1.1** | completing the square twice **soi**<br><br>**NB** $(x-6)^2-36+\ (y+4)^2-16+3=0$ **oe**<br>allow **B2** for $r$ = 7 unsupported |
| | | | *Alternatively*<br>$\pm 2a = -12$ **oe** and $\pm 2b = 8$ **oe**<br><br>$r$ = 7 not from wrong working | **M1**<br><br>**A1** | | <br><br>**NB** $r^2 = 6^2 + 4^2 - 3$ |
| | | | | **[2]** | | |
| **2** | **(b)** | | (6, –4) | **B1** | **1.1** | **FT** $(x \pm 6)^2 + (y \pm 4)^2$<br><br>**or FT** $\pm 2a = -12$ **oe** and $\pm 2b = 8$ **oe** |
| | | | | **[1]** | | |

**Exemplar responses for Q2(b)**

| Response | Mark |
| --- | --- |